



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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Course: Data Compression

Course Code: 705210502

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HUFFMAN CODING

Example 1

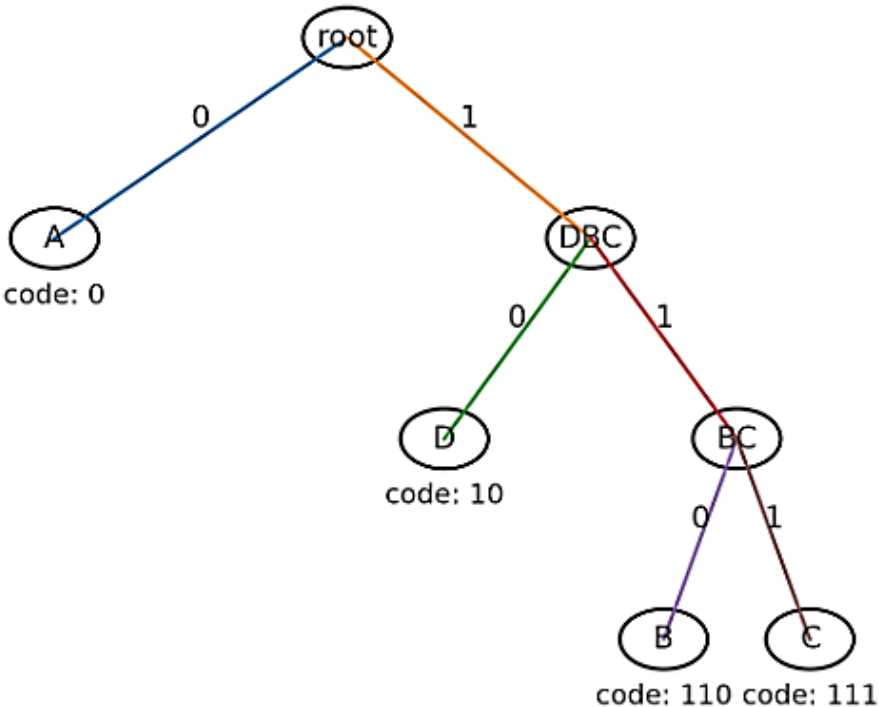
Text: "AABACD"

Symbol	Frequency	Probability	Code
A	3	$3/6 = 0.5$	0
B	1	$1/6 \approx 0.17$	10
C	1	$1/6 \approx 0.17$	110
D	1	$1/6 \approx 0.17$	111

Encoded

AABACD → 0 0 10 0 110 111 → **00100110111**

(11 bits Vs. 6×8 = 48 bits in ASCII)



So, in ASCII, you would use: 48 bits

A → 01000001 (8 bits)

B → 01000010 (8 bits)

C → 01000011 (8 bits)

D → 01000100 (8 bits)

So, 6 (Characters) × 8 (bits) = 48 bits in ASCII

But, in Huffman Coding:
the actual bit count is **11** because:

3 A's $\rightarrow 3 \times 1 \text{ bit} = 3 \text{ bits}$

1 B $\rightarrow 2 \text{ bits}$

1 C $\rightarrow 3 \text{ bits}$

1 D $\rightarrow 3 \text{ bits}$

So, $3 + 2 + 3 + 3 = \mathbf{11 \text{ bits}}$.

Comparison ASCII Vs. Huffman:

- ASCII size = $6 \times 8 = 48 \text{ bits}$
- Huffman size = **11 bits**
- Savings:

$$\frac{48 - 11}{48} \times 100\% \approx 77.08\%$$

HUFFMAN CODING

Example 2

Symbols & frequencies:

A:5, B:9, C:12, D:13, E:16, F:45

1. Count frequencies of each symbol in the input.
2. Create leaf nodes for each symbol with its frequency.
3. Build a priority queue/min-heap sorted by frequency.
- 4. Combine two smallest nodes into a new node whose frequency = sum of both.**
- 5. Repeat step 4 until only one node (root) remains → Huffman Tree.**
6. Assign codes: left branch = 0, right branch = 1.
7. Encode data using these codes.

Symbols & frequencies:

A:5, B:9, C:12, D:13, E:16, F:45

Merge A(5) + B(9) = 14(AB)

Reorder

C(12), D(13), AB(14), E(16), F(45)

Merge C(12) + D(13) = 25(CD)

Reorder

AB(14), E(16), CD(25), F(45)

Merge AB(14) + E(16) = 30(ABE)

CD(25), ABE(30), F(45)

Merge CD(25) + ABE(30) = 55(CDABE)

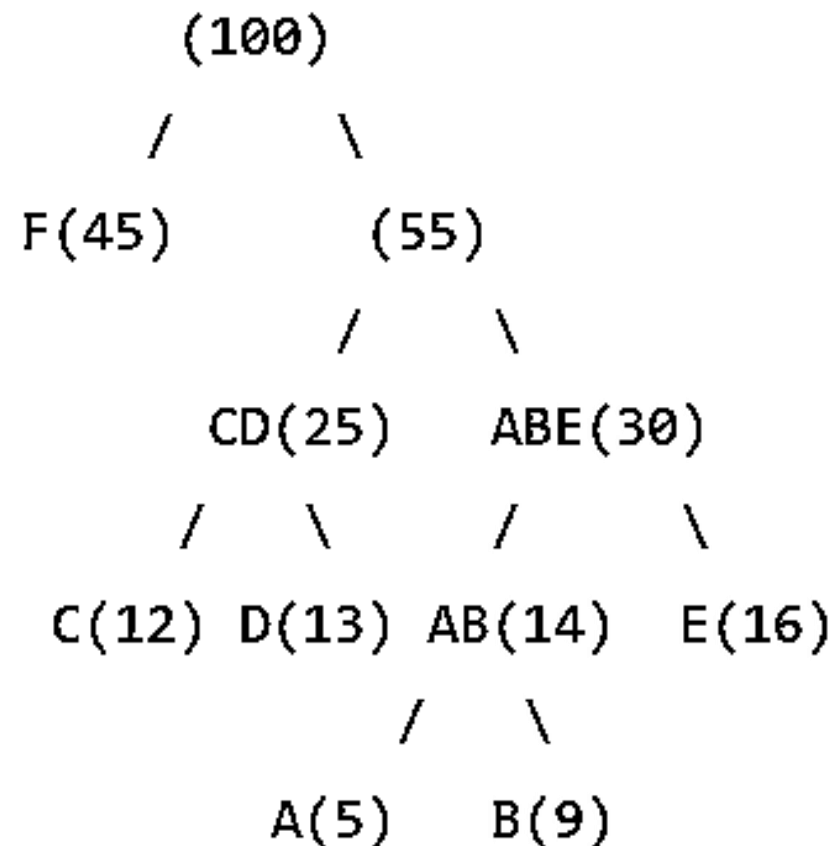
F(45), CDABE(55)

Merge F(45) + CDABE(55) = 100(root)

Now we have **one root** — tree complete.

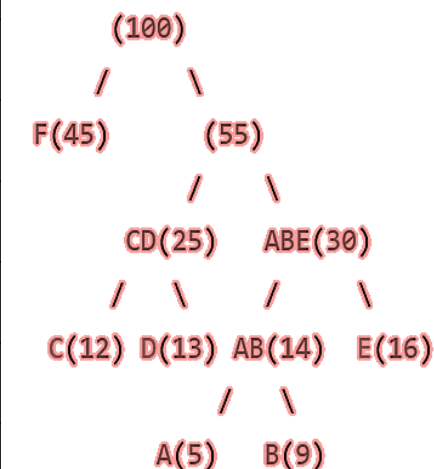
Assign 0s and 1s

Convention: Left child = 0, Right child = 1



Final (Codes: A:5, B:9, C:12, D:13, E:16, F:45)

Symbol	Frequency	Code	Code length (bits)
F	45	0	1
C	12	100	3
D	13	101	3
E	16	111	3
A	5	1100	4
B	9	1101	4



Average bits per symbol =

$$(5 \times 4) + (9 \times 4) + (12 \times 3) + (13 \times 3) + (16 \times 3) + (45 \times 1) = 224 \text{ bits for 100 symbols}$$

$$\text{Average} = 224/100 = 2.24 \text{ bits/symbol}$$

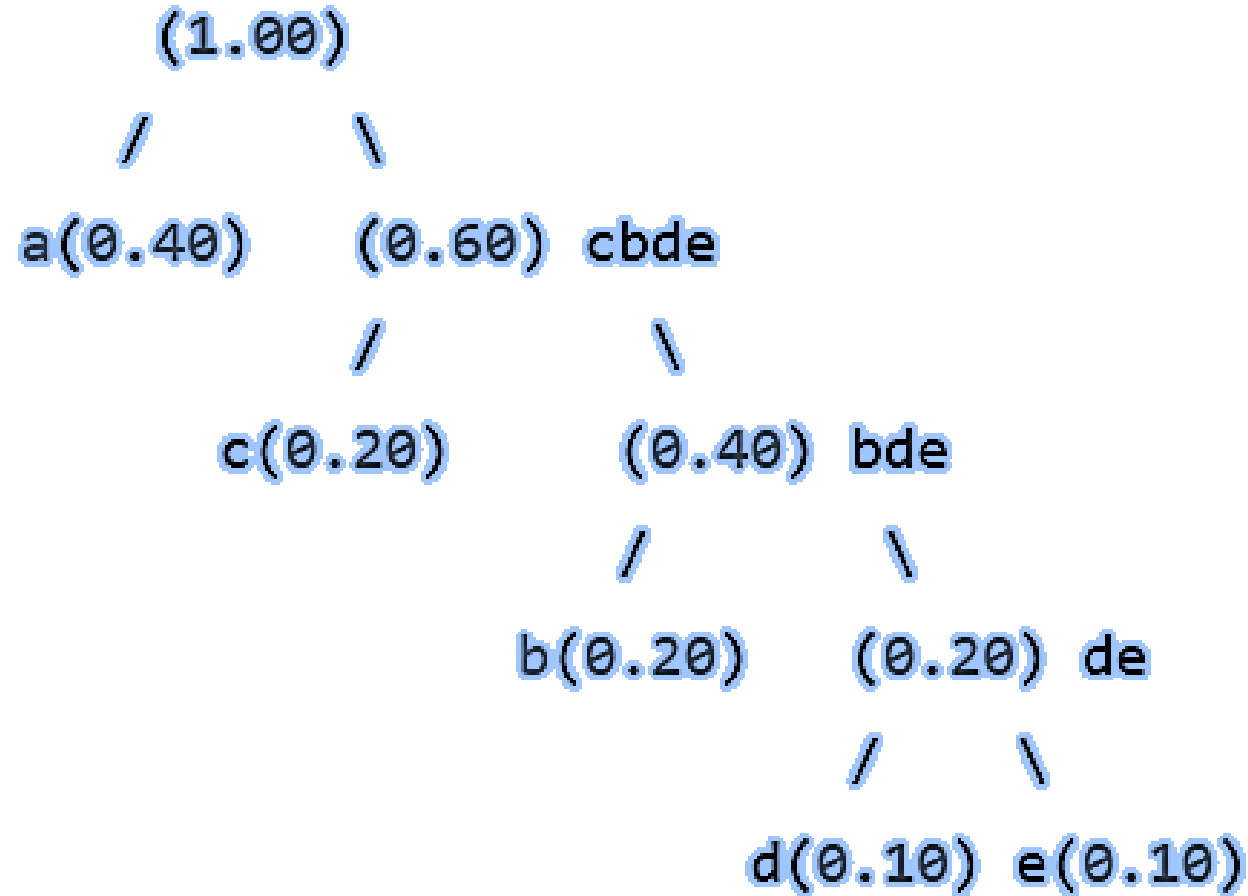
$$\frac{\text{total bits}}{\text{total symbols}} = \frac{224}{100} = 2.24 \text{ bits/symbol}$$

HUFFMAN CODING

Example 3

Symbols: {a:0.4, b:0.2, c:0.2, d:0.1, e:0.1}

a(0.40), b(0.20), c(0.20), d(0.10), e(0.10)



Final codes (from this tree):

Symbol	Probability	Code	Length (bits)
a	0.40	0	1
c	0.20	10	2
b	0.20	110	3
d	0.10	1110	4
e	0.10	1111	4

Average bits per symbol (expected length):

Compute $L = \sum p_i \cdot l_i$:

$$L = 0.4(1) + 0.2(2) + 0.2(3) + 0.1(4) + 0.1(4) = 0.4 + 0.4 + 0.6 + 0.4 + 0.4 = 2.2 \text{ bits/symbol}$$

So, on average each symbol requires 2.2 bits with this Huffman coding.

Entropy:

$$\text{Entropy } H = - \sum p_i \log_2 p_i \approx 2.122 \text{ bits/symbol}$$

Huffman average = 2.20 bits/symbol — that's close to entropy



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Thank You